Cross-Encoder Neural Reranking for Precision Evidence Selection
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ABSTRACT
First-stage dense vector retrieval maps queries and documents into isolated vector representations, losing fine-graine

1. THE TWO-STAGE RETRIEVAL PARADIGM
Bi-encoder architectures optimize for search latency at the expense of semantic precision. Because queries and pass

2. CROSS-ENCODER ARCHITECTURE & CALIBRATION

2.1 Transformer Cross-Encoder Model

We employ cross-encoder/ms-marco-MiniLM-L-6-v2, fine-tuned on the Microsoft MS-MARCO passage ranking datase

2.2 Sigmoid Relevance Score Calibration

The model outputs an unconstrained scalar logit z in (-infinity, +infinity). To provide human-interpretable relevance pro

sigma(z) = 1 / (1 + exp(-z))

This maps raw scores monotonically into the bounded interval [0.0, 1.0]. A score of 0.85 indicates strong contextual re

3. COMPARISON INTENT DETECTION & QUERY PARSING

3.1 Deterministic Intent Detection

When users ask comparative questions across multiple documents, conventional retrieval often suffers from single-do

3.2 Zero-Latency Classification

Unlike LLM-based query intent classifiers that introduce 300ms to 600ms latency, deterministic regex token matching

4. MULTI-DOCUMENT SOURCE BALANCING ALGORITHM

4.1 Source Starvation Mitigation

In multi-document comparison queries, one dense document may monopolize all top K candidate slots. To solve this,

4.2 Round-Robin Interleaving

When comparison intent is active, candidate chunks are selected via round-robin allocation:
doc1_chunk0, doc2_chunk0, doc1_chunk1, doc2_chunk1.
This ensures that both documents receive balanced representation in the prompt context presented to Gemini.

5. QUANTITATIVE RERANKING BENCHMARKS & LATENCY

5.1 Latency Profile
Cross-encoder scoring on candidate pools of size N = 15 adds an average of 185.4ms latency on modern multi-core C

5.2 Retrieval Performance Comparison
Reranking candidate pools down to top K = 5 elevates the Hit Rate@5 from 82.0% to 94.0%. Mean Reciprocal Rank (

6. ABLATION STUDY: CANDIDATE POOL SIZE (N)

We investigated the effect of varying first-stage candidate pool size N in [5, 10, 15, 20, 30]:
1. N = 5: Insufficient recall; reranker cannot recover relevant passages missed by the bi-encoder.
2. N = 15: Optimal sweet spot; achieves 96.2% candidate recall with 185.4ms CPU reranking time.
3. N = 30: Diminishing returns; marginal +0.8% recall gain accompanied by latency doubling to 370ms.

7. HARDWARE ACCELERATION & MODEL OPTIMIZATION

We benchmarked cross-encoder inference across multiple execution runtimes:
1. PyTorch CPU (FP32): 185.4ms average latency for 15 candidate pairs.
2. ONNX Runtime CPU (INT8 Quantization): 58.2ms average latency, representing a 3.2x speedup with <0.005 score
3. CUDA GPU (FP16): 22.1ms average latency, optimal for high-throughput batch evaluation.

8. ERROR ANALYSIS OF RERANKING FAILURES

Cross-encoder reranking failures primarily occur under two conditions:
1. Candidate Truncation: Long technical definitions exceeding 1000 characters have critical conclusions clipped.
2. Semantic False Friends: Passages with high lexical overlap but contradictory semantic assertions can receive inflat